

SYSTEM FOUNDRY / PERSONAL MVP

RenewalRadar

A local-first renewal queue that turns scattered subscriptions into a visible annual cost.

PRODUCT STATUS	AUTOMATED TESTS	ONE-TIME PRICE
PASS	8 / 8	KRW 9.9K
Implemented and verified	Zero failures	Personal license

01 / PROJECT OVERVIEW

Project overview

RenewalRadar lets an individual register recurring digital services, see what will charge within seven days, and compare monthly-equivalent, annual, and canceled-savings totals without connecting a bank account.

02 / TARGET CUSTOMER

Target customer

People paying for several music, creative, cloud, education, or web subscriptions who want a lightweight view of renewal dates and fixed annual spending.

03 / PROBLEM

Problem

Small recurring payments are distributed across stores and email receipts. The next charge is easy to miss, while a monthly price hides the annual commitment and makes cancellation decisions harder.

PRODUCT

Three focused jobs

04 / CORE FEATURES

Core features

The MVP implements one complete subscription review loop:

01	Validated subscription intake Capture service, category, amount, billing cycle, next renewal, and status with strict limits.
02	Renewal urgency queue Order today's, seven-day, later, paused, and canceled services around the next useful decision.
03	True cost and savings totals Calculate monthly-equivalent and annual cost, then show annual savings from canceled services.

05 / HOW TO USE

How to use

Register each active subscription with its next charge date. Review today's and seven-day items before renewal. Change an unused service to canceled so active cost falls and the avoided annual cost remains visible.

06 / SYSTEM ARCHITECTURE

System architecture

A static browser UI calls pure validation, urgency, and cost functions and persists state only in localStorage. Node tests exercise the same module. No account, backend, database, analytics, bank access, or external API is used.

Browser UI	Tracker module	Local storage	Node tests
HTML + CSS	Rules + totals	Device-only data	Pure functions

07 / TECHNOLOGY

Technology

Static HTML, responsive CSS, vanilla JavaScript, localStorage, bundled Noto Sans KR fonts, and the built-in Node.js test runner. Runtime dependencies: zero. Paid APIs: zero.

VALIDATED INTERFACE

Actual running screen

Captured from headless Chromium after a new audiobook subscription was validated and added.

The screenshot displays the RenewalRadar 'PERSONAL RENEWAL DESK'. At the top, a message states: '구독을 등록하고 경신 쿠폰 업데이트했습니다.' (Registered subscriptions and updated renewal coupons). The main header features the slogan '조용히 새는 돈, 결제 전에 잡자.' (Quietly leaking money, let's catch it before payment) and a sub-header '이번 주 결제와 연간 고정비를 한 화면에서 관리하세요.' (Manage this week's payment and annual fixed fee on one screen).

Four summary cards are shown: '활성 구독' (Active Subscriptions) with a count of 5, '7일 내 결제' (Payment due within 7 days) with a count of 4, '월 환산 비용' (Monthly Equivalent Cost) of ₩38,200, and '연간 총액' (Annual Total) of ₩458,400 (with a note '계좌 월금 ₩106,600').

The 'RENEWAL QUEUE' section, titled '다가오는 결제' (Upcoming Payments), lists five subscriptions with their renewal dates and costs:

Subscription	Renewal Date	Next Payment	Status
Music Cloud	2026-08-05	₩10,900	이용 중 (In Use)
Audio Book	2026-08-06	₩7,900	이용 중 (In Use)
Edit Pro	2026-08-07	₩13,000	이용 중 (In Use)
My Domain	2026-08-12	₩18,000	이용 중 (In Use)
Drive Box	2026-08-30	₩4,900	이용 중 (In Use)

Below this, a '해지' (Cancel) section shows 'Course Club' with a renewal date of 2026-08-02 and a cost of ₩8,900.

On the right, the 'NEW SUBSCRIPTION' section, titled '구독 등록' (Register Subscription), includes a description: '결제 예정일을 넣으면 연간 비용까지 계산합니다.' (If you enter the payment due date, the annual cost will be calculated). It features input fields for '서비스명' (Service Name) with '예: Audio Book', '분류' (Category) with '예: 오디오북', '결제 금액' (Payment Amount) with '7900', '결제 주기' (Payment Cycle) with '매월' (Monthly), '다음 경신일' (Next Renewal Date) with '08/06/2026', and '상태' (Status) with '이용 중' (In Use). A '구독 등록' (Register Subscription) button is at the bottom.

A 'LOCAL FIRST' notice at the bottom right states: '계정 정보나 결제 개인정보는 저장하지 않습니다.' (We do not store account information or payment personal information).

08 / RUNNING SCREEN

Running screen

The verified screen shows five active subscriptions, four due within seven days, KRW 38,200 monthly-equivalent cost, and KRW 458,400 annual cost after adding a KRW 7,900 audiobook plan. Canceling that plan returned active totals to baseline and raised canceled annual savings to KRW 201,600.

VERIFICATION

Evidence, not assumptions

09 / TEST RESULTS

Test results

Every required check was executed against the generated project.

Check	Result	Evidence
Dependency installation	PASS	npm install completed; zero runtime dependencies
Application start	PASS	README HTTP server returned status 200
Core workflow	PASS	Active 4 to 5 to 4; savings to KRW 201.6K
Automated tests	8 / 8	Validation, cycles, urgency, totals, status
README commands	PASS	Install, test, and start commands executed
Credential scan	PASS	No secret patterns and no .env file
PDF	PASS	Generated, parsed, rendered, visually inspected

10 / SECURITY AND LIMITATIONS

Security and limitations

RenewalRadar sends no data outside the browser and never asks for account credentials, card numbers, or bank access. Text, amount, cycle, date, and status are validated; displayed text is HTML-escaped. Device-local data is unencrypted and disappears when browser storage is cleared. Renewal dates and prices require manual updates, and the MVP does not send notifications or cancel services automatically.

GO TO MARKET

One avoided renewal pays for it

11 / PRICING AND REVENUE MODEL

Pricing and revenue model

Sell a downloadable personal license for KRW 9,900 as a one-time purchase. The price is below many single monthly renewals and the zero-hosting design keeps ongoing cost low. Updates can be included for one major version.

12 / FIRST CUSTOMER ACQUISITION

First customer acquisition

Recruit twenty users from Korean personal-finance, productivity, student, and creator communities. Ask each user to enter five subscriptions, complete one renewal review, and report any canceled annual value. Convert with the visible before-and-after annual cost, not a vague budgeting promise.

13 / IMPROVEMENT PLAN

Improvement plan

Priorities after observing real renewal routines:

- JSON backup and restore for device migration.
- Calendar export for optional user-controlled reminders.
- Price-change history and renewal review notes.
- Private encrypted sync only if multi-device demand is proven.

PRODUCT BOUNDARY

RenewalRadar should remain a lightweight review tool. It should not request banking credentials, cancel services without the user's direct action, or present manually entered costs as verified account balances.

ASSESSMENT

Overall evaluation

14 / OVERALL SCORE

Overall score

Six dimensions are scored out of 10. A higher maintenance score means easier maintenance.

Dimension	Score	Rationale
Customer problem strength	8/10	Unnoticed renewals create recurring and measurable personal cost.
One-day completeness	9/10	The register, prioritize, total, and cancel loop is browser-tested.
Sales potential	8/10	One avoided renewal can clearly justify the one-time price.
Maintenance difficulty	9/10	Zero runtime dependencies and no backend keep maintenance light.
Differentiation	7/10	Annual savings visibility helps, though subscription trackers exist.
Scalability	8/10	Backups, calendar export, history, and sync offer expansion paths.

OVERALL

8.2 / 10

Proceed to a twenty-user renewal review trial.

Verdict: technically complete and visually polished for a personal MVP. The next useful evidence is whether users return before renewal dates and record enough avoided cost to justify purchase.